

Department: Physics Department, University of the Balearic Islands (UIB)

Position: PostDoc

Affiliation: IFISC (Institute for Cross-Disciplinary Physics and Complex Systems) and UIB

Education

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| Feb 2026 – present | ■ Postdoctoral Researcher, IFISC and University of the Balearic Islands |
| Jan 2021 – Feb 2026 | ■ PhD Student, IFISC and University of the Balearic Islands in Physics.
Supervisor: <i>Tobias Galla, Lucas Lacasa</i>
Thesis title: <i>Complex systems with time-varying interactions.</i>
Grade: <i>Cum laude</i> |
| Apr 2024 – Jun 2024 | ■ Visiting Student, Queen Mary University of London.
Supervisor: <i>Vito Latora</i> |
| Sep 2017 – Jan 2020 | ■ M.Sc. in Physics, Sapienza University of Rome (Theoretical Physics).
Thesis title: <i>Degree-ordered percolation on complex networks.</i>
Supervisor: <i>Claudio Castellano</i>
Grade: <i>110/110 with honors</i> |
| Sep 2014 – Sep 2017 | ■ B.Sc. in Physics, Sapienza University of Rome.
Thesis title: <i>Analysis of chaotic systems with the Kolmogorov–Sinai entropy and Lyapunov exponents.</i>
Supervisor: <i>Vittorio Loreto</i>
Grade: <i>110/110 with honors</i> |

My research lies at the intersection of dynamical systems and network science. During my PhD, I have focused on the study of graph dynamical systems, where the object evolving in time is a network rather than a point in phase space. In particular, I investigate how concepts from dynamical systems theory—such as sensitivity to initial conditions and Lyapunov exponents—can be extended to temporal networks.

Research Interests

Temporal networks, Chaos and nonlinear systems, Stochastic processes, Opinion dynamics Models

Publication List

- 1 Predictability of temporal network dynamics in normal ageing and brain pathology. (2026). *Journal of Complex Networks*, 14(1), cnafo58.  doi:10.1093/comnet/cnaf058
- 2 Characterizing the dynamics of unlabeled temporal networks. (2025). *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 35(5), 053122.  doi:10.1063/5.0253870
- 3 Time-varying ecological interactions characterise equilibrium and stability. (2025). *arXiv preprint arXiv:2506.22123*.

- 4 Noisy voter models in switching environments. (2023). *Phys. Rev. E*, 108, 044301.
[doi:10.1103/PhysRevE.108.044301](https://doi.org/10.1103/PhysRevE.108.044301)
- 5 Lyapunov exponents for temporal networks. (2023). *Phys. Rev. E*, 107, 044305.
[doi:10.1103/PhysRevE.107.044305](https://doi.org/10.1103/PhysRevE.107.044305)
- 6 Degree-ordered-percolation on uncorrelated networks. (2020). *Journal of Statistical Mechanics: Theory and Experiment*, 2020(11), 113401. [doi:10.1088/1742-5468/abc1da](https://doi.org/10.1088/1742-5468/abc1da)

Conferences

- **NetSci2020** (First online conference): Poster with the title "Degree-Ordered-Percolation on complex networks";
- **FisEs Zaragoza, Spain** Poster with the title "Noisy Voter Model with time-varying influencers".
- **GAINs, Vienna** Attendance of summer school with the title 'Disintegration of societies' organized by the Santa Fe Institute in collaboration with Complexity Science Hub Vienna and other European institutions.
- **CCS22, Palma** Talk with the title "Noisy Voter Model with time-varying influencers".
- **NetSci23, Vienna** Talk with the title "Lyapunov Exponents for Temporal Networks".
- **NetPLACE@NetSci23, Vienna** organiser of NetPLACE@NetSci23
- **DD23, Naples** Talk with the title "Lyapunov Exponents for Temporal Networks".
- **Winter Workshop on Complex Systems 2024, Vall d'en Bas, Spain:** Worked on a small project using bees' proximity data.
- **1st NetPlace workshop on Networks & Physics (London, May 2025)** organiser.
- **NetSci25, Maastricht** Poster at TENET satellite.
- **Complexity72h 2025, Madrid** Preprint with the title: Time-varying ecological interactions characterise equilibrium and stability.
- **NetSciX26, Auckland** Talk with the title "Characterizing the dynamics of unlabeled temporal networks".
- **NetSciX26, Auckland** Moderator of the panel "Beyond Isolation: Getting Connected in Network Science" organized by NetPlace.

Outreach

- Ex organizer of NetPLACE, a community for the early-career researchers in Network Science.
- 11F International Day of Women and Girls in Science (in Spanish)

Teaching

- Physics I for the bachelor degree in Chemistry at University of the Balearic Islands (UIB): tutorial sessions (2023-2024).

Skills

Languages	📖	Italian (Mother Tongue), English (C1), Spanish (Fluent), French and German (Basic)
Coding	📖	C, Python, $\text{\LaTeX}$, Matlab, Mathematica
Theoretical methods	📖	Stochastic Processes, Path Integrals, Network Theory, Chaos Theory
Further knowledge	📖	Online courses: • "Game theory" • "Machine Learning with Python: A Practical Introduction"
Other interests	📖	Outside my main research trajectory, I am a passionate indoor and outdoor climber and hold the certification of Indoor Climbing Instructor.

References

Dr Claudio Castellano

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